

1、

$$(a) 11^{644} \bmod 645 \quad 11^{644} = 11^{2^9 + 2^7 + 2^2} \\ = 11^{2^9} \times 11^{2^7} \times 11^{2^2}$$

$$\begin{array}{r} 2 \overline{) 644} \quad 0 \\ 2 \overline{) 322} \quad 0 \\ 2 \overline{) 161} \quad 1 \\ 2 \overline{) 80} \quad 0 \\ 2 \overline{) 40} \quad 0 \\ 2 \overline{) 20} \quad 0 \\ 2 \overline{) 10} \quad 0 \\ 2 \overline{) 5} \quad 1 \\ 2 \overline{) 1} \quad 0 \\ \quad \quad 1 \end{array}$$

$$644_{10} = 1010000100_2 \\ \quad \quad \quad \overline{2^9} \quad \overline{2^7} \quad \quad \overline{2^2}$$

$$11^{2^0} \bmod 645 = 11^1 \bmod 645 = 11 \bmod 645 = 11$$

$$11^{2^1} \bmod 645 = 11^2 \bmod 645 = 121 \bmod 645 = 121$$

$$\checkmark 11^{2^2} \bmod 645 = 11^4 \bmod 645 = 121^2 \bmod 645 = 451$$

$$11^{2^3} \bmod 645 = 11^8 \bmod 645 = 451^2 \bmod 645 = 226$$

$$11^{2^4} \bmod 645 = 11^{16} \bmod 645 = 226^2 \bmod 645 = 121$$

$$11^{2^5} \bmod 645 = 11^{32} \bmod 645 = 121^2 \bmod 645 = 451$$

$$11^{2^6} \bmod 645 = 11^{64} \bmod 645 = 451^2 \bmod 645 = 226$$

$$\checkmark 11^{2^7} \bmod 645 = 11^{128} \bmod 645 = 226^2 \bmod 645 = 121$$

$$11^{2^8} \bmod 645 = 11^{256} \bmod 645 = 121^2 \bmod 645 = 451$$

$$\checkmark 11^{2^9} \bmod 645 = 11^{512} \bmod 645 = 451^2 \bmod 645 = 226$$

$$3^{644} \bmod 645 = (11^{2^9} \times 11^{2^7} \times 11^{2^2}) \bmod 645$$

$$= (226 \times 121 \times 451) \bmod 645 = 1 \neq \checkmark$$

$$(b) 3^{2003} \bmod 99$$

$$\begin{array}{r} 2 \overline{) 2003} - 1 \\ 2 \overline{) 1001} - 1 \\ 2 \overline{) 500} - 0 \\ 2 \overline{) 250} - 0 \\ 2 \overline{) 125} - 1 \\ 2 \overline{) 62} - 0 \\ 2 \overline{) 31} - 1 \\ 2 \overline{) 15} - 1 \\ 2 \overline{) 7} - 1 \\ 2 \overline{) 3} - 1 \\ 1 \end{array}$$

$$2003_{10} = \frac{1111}{10} \frac{101001}{8764} \frac{1}{10} \frac{1}{2}$$

$$3^{2003} = 3^{2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^4 + 2^1 + 2^0}$$

$$\checkmark 3^{2^0} \bmod 99 = 3$$

$$\checkmark 3^{2^1} \bmod 99 = 9 \bmod 99 = 9$$

$$3^{2^2} \bmod 99 = 9^2 \bmod 99 = 81$$

$$3^{2^3} \bmod 99 = 81^2 \bmod 99 = 27$$

$$\checkmark 3^{2^4} \bmod 99 = 27^2 \bmod 99 = 36$$

$$3^{2^5} \bmod 99 = 36^2 \bmod 99 = 9$$

$$\checkmark 3^{2^6} \bmod 99 = 9^2 \bmod 99 = 81$$

$$\checkmark 3^{2^7} \bmod 99 = 81^2 \bmod 99 = 27$$

$$\checkmark 3^{2^8} \bmod 99 = 27^2 \bmod 99 = 36$$

$$\checkmark 3^{2^9} \bmod 99 = 36^2 \bmod 99 = 9$$

$$\checkmark 3^{2^{10}} \bmod 99 = 9^2 \bmod 99 = 81$$

$$\begin{aligned} 3^{2003} \bmod 99 &= (3 \times 9 \times 36 \times 81 \times 27 \times 36 \times 9 \times 81) \bmod 99 \\ &= 55788550416 \bmod 99 = 27 \quad \checkmark \end{aligned}$$

$$(c) 123^{1001} \bmod 101$$

By Fermat little theorem, $(123^{100})^k \equiv 1 \pmod{101}$, $k \in \mathbb{Z}^+$

$$123^{1001} = 123^{100 \times 10 + 1} = (123^{100})^{10} 123 \equiv 1^{10} \cdot 123 \equiv 22 \pmod{101}$$

$$123^{1001} \bmod 101 = 22 \quad \checkmark$$

$$(d) 7^{121} \bmod 13$$

By Fermat's little theorem,

$$(7^{12})^k \equiv 1 \pmod{13}, k \in \mathbb{Z}^+$$

$$7^{121} = 7^{12 \times 10 + 1}$$

$$7^{121} \bmod 13$$

$$= (7^{12})^{10} 7 \bmod 13$$

$$= [(7^{12})^{10} \bmod 13] \times (7 \bmod 13) \bmod 13$$

$$= (1 \times 7) \bmod 13 = 7 \checkmark$$

$$(f) \gcd(1529, 14039)$$

$$\begin{array}{r|l} 5 & 1529 \mid 14039 \mid 9 \\ & 1390 \mid 3761 \\ \hline & 139 \mid 278 \mid 2 \\ & 278 \mid 0 \end{array}$$

$$\gcd(1529, 14039) = 139 \checkmark$$

$$(e) 23^{1002} \bmod 41$$

By Fermat's little theorem,

$$(23^{40})^k \equiv 1 \pmod{41}, k \in \mathbb{Z}^+$$

$$23^{1002} = (23^{40})^{25+2}$$

$$23^{1002} \bmod 41$$

$$= [(23^{40})^{25} \bmod 41] \times (23^2 \bmod 41) \bmod 41$$

$$= (1 \times 37) \bmod 41 = 37 \checkmark$$

$$(g) \gcd(1111, 0)$$

~~No definition.~~

1111

Definition:

Let a, b be integers, not both zero.

The largest integers d such that $d \mid a$ and also $d \mid b$ is called the $\gcd$ of a and b .

And so, 0 可被任何數整除。所以

$$\gcd(1111, 0) = 1111$$

2.

(a) $97644_{10} \rightarrow \text{BIN}$ (b) $10101101_2 \rightarrow \text{DEC}$

$$\begin{array}{r}
 2 \overline{) 97644} - 0 \\
 2 \overline{) 48822} - 0 \\
 2 \overline{) 24411} - 1 \\
 2 \overline{) 12205} - 1 \\
 2 \overline{) 6102} - 0 \\
 2 \overline{) 3051} - 1 \\
 2 \overline{) 1525} - 1 \\
 2 \overline{) 762} - 0 \\
 2 \overline{) 381} - 1 \\
 2 \overline{) 190} - 0 \\
 2 \overline{) 95} - 1 \\
 2 \overline{) 47} - 1 \\
 2 \overline{) 23} - 1 \\
 2 \overline{) 11} - 1 \\
 2 \overline{) 5} - 1 \\
 2 \overline{) 2} - 0
 \end{array}$$

$$97644_{10}$$

$$= \underline{10111101010100}_2 \checkmark$$

$$\begin{array}{ccccccc}
 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \\
 \overline{2^9} & & \overline{2^7} & & \overline{2^5} & \overline{2^4} & & \overline{2^2} & & \overline{2^0}
 \end{array}$$

$$= 2^9 + 2^7 + 2^5 + 2^4 + 2^2 + 2^0$$

$$= 1 + 4 + 16 + 32 + 128 + 512$$

$$= 693_{10} \checkmark$$

(c) $423_8 \rightarrow \text{BIN}$

$$\begin{array}{ccc}
 \begin{array}{c} 4 \\ \hline T \end{array} & \begin{array}{c} 2 \\ \hline T \end{array} & \begin{array}{c} 3 \\ \hline T \end{array} \\
 100 & 010 & 011
 \end{array}$$

$$423_8 = \underline{100010011}_2 \checkmark$$

(d) $1010101010_2 \rightarrow \text{OCT}$

$$\begin{array}{cccc}
 \begin{array}{c} 101 \\ \hline 5 \end{array} & \begin{array}{c} 010 \\ \hline 2 \end{array} & \begin{array}{c} 101 \\ \hline 5 \end{array} & \begin{array}{c} 010 \\ \hline 2 \end{array}
 \end{array}$$

$$1010101010_2 = 5252_8 \checkmark$$

A 10

B 11

C 12

D 13

E 14

F 15

(e) $135AB_{16} \rightarrow \text{OCT}$

$$\begin{array}{cccccc}
 \begin{array}{c} 1 \\ \hline 000 \end{array} & \begin{array}{c} 3 \\ \hline 001 \end{array} & \begin{array}{c} 5 \\ \hline 010 \end{array} & \begin{array}{c} A \\ \hline 1010 \end{array} & \begin{array}{c} B \\ \hline 1011 \end{array} \\
 0 & 2 & 3 & 2 & 6 & 5 & 3
 \end{array}$$

$$135AB_{16} = 232653_8 \checkmark$$

(f) $BADFACED_{16} \rightarrow \text{OCT}$

$$\begin{array}{ccccccccc}
 \begin{array}{c} B \\ \hline 1011 \end{array} & \begin{array}{c} A \\ \hline 1010 \end{array} & \begin{array}{c} D \\ \hline 1101 \end{array} & \begin{array}{c} F \\ \hline 1111 \end{array} & \begin{array}{c} A \\ \hline 1010 \end{array} & \begin{array}{c} C \\ \hline 1100 \end{array} & \begin{array}{c} E \\ \hline 1110 \end{array} & \begin{array}{c} D \\ \hline 1101 \end{array} \\
 2 & 7 & 6 & 7 & 2 & 6 & 3 & 5
 \end{array}$$

$$BADFACED_{16} = 27267726355_8 \checkmark$$

(g) $1011011101_2 \rightarrow \text{OCT}$

$$\begin{array}{cccc}
 \begin{array}{c} 101 \\ \hline 5 \end{array} & \begin{array}{c} 101 \\ \hline 5 \end{array} & \begin{array}{c} 111 \\ \hline 7 \end{array} & \begin{array}{c} 011 \\ \hline 3 \end{array}
 \end{array}$$

$$1011011101_2 = 5573_8 \checkmark$$

3.

(a)

$$\begin{array}{r} 100 \ 011 \\ + 111 \ 011 \\ \hline 1011 \ 1110 \end{array} \quad \begin{array}{r} 71 \\ \times 119 \\ \hline 639 \\ 71 \\ \hline 8449 \end{array}$$

$$1000111_2 = 71$$

$$1110111_2 = 119$$

$$8449 = 8192 + 257$$

$$(2^{13}) \quad \sim \quad 256 + 1$$

$$(2^8) + 2^0$$

$$8449_{10} = 10000 \ 00000000$$

$$\text{Ans: sum} = 1011110$$

$$\text{product} = 10000 \ 00000000$$

(b)

$$\begin{array}{r} 112 \\ + 210 \\ \hline 1022 \end{array} \quad \begin{array}{l} \frac{1}{3^2} \frac{1}{3^1} \frac{2}{3^0} = 9 + 3 + 2 = 14 \\ \frac{2}{3^2} \frac{1}{3^1} \frac{0}{3^0} = 18 + 3 + 0 = 21 \end{array}$$

$$\begin{array}{r} 14 \\ \times 21 \\ \hline 14 \\ 28 \\ \hline 294 \end{array}$$

$$\begin{array}{r} 3 \overline{) 294} - 0 \\ 3 \overline{) 98} - 2 \\ 3 \overline{) 32} - 2 \\ 3 \overline{) 10} - 1 \\ 3 \overline{) 3} - 0 \\ 1 \end{array}$$

Ans:

$$\text{sum} = 1022$$

$$\text{product} = 101220$$

(c)

$$\begin{array}{r} 763 \\ + 147 \\ \hline 1132 \end{array} \quad \begin{array}{r} 763 \\ \times 147 \\ \hline 15341 \\ 3052 \\ 763 \\ \hline 134361 \end{array}$$

$$\text{Ans: sum} = 1132$$

$$\text{product} = 134361$$

$$763_8 = 499_{10}$$

$$147_8 = 103_{10}$$

$$499 + 103 = 602 = 1132_8$$

$$499 \times 103 = 51397 = 144305_8$$

(d)

$$\begin{array}{r} 1AE \\ + BBC \\ \hline D6A \end{array}$$

$$\begin{array}{r} 1AE \\ \times BBC \\ \hline 1428 \\ 127A \\ 127A \\ \hline 13B5C8 \end{array}$$

$$\text{Ans: sum} = D6A$$

$$\text{product} = 13B5C8$$

$$A \ 10$$

$$B \ 11$$

$$C \ 12$$

$$D \ 13$$

$$E \ 14$$

$$F \ 15$$

4. $a \equiv 4 \pmod{13}$
 $b \equiv 9 \pmod{13}$, find integer c with $0 \leq c \leq 12$

(a) $c \equiv 11b \pmod{13}$

$$b \equiv 9 \pmod{13}$$

$$11b \equiv 99 \pmod{13}$$

$$c \equiv 11b \pmod{13}$$

$$\equiv 99 \pmod{13}$$

$$\equiv 8 \pmod{13}, \underline{c=8}$$

(c) $c \equiv a^3 - b^3 \pmod{13}$

$$a^3 \equiv 4^3 \pmod{13}$$

$$\equiv 64 \pmod{13}$$

$$b^3 \equiv 9^3 \pmod{13}$$

$$\equiv 729 \pmod{13}$$

$$c \equiv a^3 - b^3 \pmod{13}$$

$$\equiv 64 - 729 \pmod{13}$$

$$\equiv -665 \pmod{13}$$

$$\equiv -2 \pmod{13}$$

$$\equiv \underline{11} \pmod{13}$$

$$\begin{array}{r} 51 \\ 13 \overline{) 665} \\ \underline{65} \\ 15 \\ \underline{13} \\ 2 \end{array}$$

$$a < 0,$$

$$a \bmod b = b - \text{餘}$$

$$13 - 2 = 11$$

5.

(a) $(177 \bmod 3 | \times 270 \bmod 3 |) \bmod 3 |$

$$= (22^2) \bmod 3 |$$

$$= 19_{\#} \checkmark$$

(c) $(32^3 \bmod 13)^2 \bmod 11$

$$= (32 \bmod 13)^3 \bmod 13)^2 \bmod 11$$

$$= (216 \bmod 13)^2 \bmod 11$$

$$= 64 \bmod 11 = 9_{\#} \checkmark$$

(b) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$

$$= (5 + 8) \bmod 23$$

$$= 13_{\#} \checkmark$$

(d) $(99^2 \bmod 32)^3 \bmod 15$

$$= ((99 \bmod 32)^2 \bmod 32)^3 \bmod 15$$

$$= (9 \bmod 32)^3 \bmod 15$$

$$= 729 \bmod 15 = 9_{\#} \checkmark$$

6.

(a) 117, 213

$$213 = 1 \times 117 + 96$$

$$117 = 1 \times 96 + 21$$

$$96 = 4 \times 21 + 12$$

$$21 = 1 \times 12 + 9$$

$$12 = 1 \times 9 + 3$$

$$9 = 3 \times \underline{3} + 0$$

$$\gcd(117, 213) = 3 \#$$

$$3 = 12 - 9$$

$$= 1 \times 12 + (-1) \times (21 - 12)$$

$$= 2 \times 12 + (-21)$$

$$= 2(96 - 4 \times 21) + (-21)$$

$$= 2 \times 96 - 9 \times 21$$

$$= 2 \times 96 - 9 \times (117 - 96)$$

$$= 11 \times 96 - 9 \times 117$$

$$= 11 \times (213 - 117) - 9 \times 117$$

$$= \underline{11 \times 213 - 20 \times 117} \# \checkmark$$

(b) 124, 323

$$323 = 2 \times 124 + 75$$

$$124 = 1 \times 75 + 49$$

$$75 = 1 \times 49 + 26$$

$$49 = 1 \times 26 + 23$$

$$26 = 1 \times 23 + 3$$

$$23 = 7 \times 3 + 2$$

$$3 = 1 \times 2 + 1$$

$$2 = 2 \times \underline{1} + 0$$

$$\gcd(124, 323) = 1$$

$$1 = 3 - 2$$

$$= 3 - (23 - 7 \times 3)$$

$$= 8 \times 3 - 23$$

$$= 8 \times (26 - 23) - 23$$

$$= 8 \times 26 - 9 \times 23$$

$$= 8 \times 26 - 9 \times (49 - 26)$$

$$= 17 \times 26 - 9 \times 49$$

$$= 17 \times (75 - 49) - 9 \times 49$$

$$= 17 \times 75 - 26 \times 49$$

$$= 17 \times 75 - 26 \times (124 - 75)$$

$$= 43 \times 75 - 26 \times 124$$

$$= 43(323 - 2 \times 124) - 26 \times 124$$

$$= \underline{43 \times 323 - 112 \times 124} \# \checkmark$$

$$7. ax \equiv b \pmod{m}$$

$$(a) 4x \equiv 5 \pmod{9}$$

$$7 \times 4x \equiv 7 \times 5 \pmod{9}$$

$$9 = 2 \times 4 + 1 \quad 1 = 9 - 2 \times 4$$

$$x \equiv 35 \pmod{9}$$

$$4 = 4 \times 1 + 0 \quad = 1 \times 9 + (-2) \times 4$$

$$\equiv 8$$

$$\gcd(4, 9) = 1 \quad \text{inverse} = -2 + k$$

$$8 + 9k, k \in \mathbb{Z}$$

$$-2 + 9 = 7$$

$$(b) 34x \equiv 77 \pmod{89}$$

$$89 = 2 \times 34 + 21 \quad 1 = 3 - 2$$

$$34 = 1 \times 21 + 13 \quad = 3 - (5 - 3)$$

$$21 = 1 \times 13 + 8 \quad = 2 \times 3 - 5$$

$$13 = 1 \times 8 + 5 \quad = 2 \times (8 - 5) - 5$$

$$8 = 1 \times 5 + 3 \quad = 2 \times 8 - 3 \times 5$$

$$5 = 1 \times 3 + 2 \quad = 2 \times 8 - 3 \times (13 - 8)$$

$$3 = 1 \times 2 + 1 \quad = 5 \times 8 - 3 \times 13$$

$$2 = 2 \times 1 + 0 \quad = 5 \times (21 - 13) - 3 \times 13$$

$$\gcd(34, 89) = 1 \quad = 5 \times 21 - 8 \times 13$$

$$-34 + 89 = 55 \quad = 5 \times 21 - 8 \times (34 - 21)$$

$$55 \times 34x \equiv 55 \times 77 \pmod{89} \quad = 13 \times 21 - 8 \times 34$$

$$x \equiv 4235 \pmod{89} \quad = 13 \times (89 - 2 \times 34) - 8 \times 34$$

$$x \equiv 52 \pmod{89} \quad = 13 \times 89 - 34 \times 34$$

$$= 13 \times 89 + (-34) \times 34$$

$$52 + 89k, k \in \mathbb{Z}$$

$$\text{inverse} = -34 + k$$

$$(c) 15x^2 + 19x \equiv 5 \pmod{11}$$

$$15x^2 + 19x + 6 = 0$$

$$< \text{Hint: } 15x^2 + 19x + 6 \equiv 0 \pmod{11} >$$

$$\begin{array}{r} 5x \quad \times \quad 3 \\ 3x \quad \times \quad 2 \\ \hline (5x+3)(3x+2) = 0 \end{array}$$

$$\underbrace{(5x+3)}_{\textcircled{1}} \underbrace{(3x+2)}_{\textcircled{2}} \equiv 0 \pmod{11}$$

①

$$5x+3 \equiv 0 \pmod{11}$$

$$5x + \cancel{3+8} \equiv 0+8 \pmod{11}$$

$$5x \equiv 8 \pmod{11}$$

$$11 = 2 \times 5 + 1 \quad 1 = 11 - 2 \times 5$$

$$5 = 5 \times 1 + 0 \quad = 11 + (-2) \times 5$$

$$\gcd(5, 11) = 1 \quad \text{inverse: } -2+k$$

$$-2+11 = 9$$

$$9 \times 5x \equiv 9 \times 8 \pmod{11}$$

$$x \equiv 72 \pmod{11}$$

$$x \equiv 6 \pmod{11}$$

②

$$3x+2 \equiv 0 \pmod{11}$$

$$3x + \cancel{2+9} \equiv 0+9 \pmod{11}$$

$$3x \equiv 9 \pmod{11}$$

$$11 = 3 \times 3 + 2 \quad 1 = 3 - 2$$

$$3 = 1 \times 2 + 1 \quad = 3 - (1 - 3 \times 3)$$

$$2 = 2 \times 1 + 0 \quad = 4 \times 3 - 11$$

$$\gcd(3, 11) = 1 \quad \text{inverse: } 4+k$$

$$4+11 = 15$$

$$15 \times 3x \equiv 15 \times 9 \pmod{11}$$

$$x \equiv 135 \pmod{11}$$

$$\equiv 3 \pmod{11}$$

$$\text{Ans: } x = 3 \text{ or } 6$$

$$(d) x \equiv 5 \pmod{6}$$

$$x \equiv 3 \pmod{10}, \text{ find all solution.}$$

$$x \equiv 8 \pmod{15}$$

$$x = 6t + 5, t \in \mathbb{Z}$$

$$6t + 5 \equiv 8 \pmod{15}$$

$$6t \equiv 3 \pmod{15} \rightarrow 2t \equiv 1 \pmod{5}$$

$$\gcd(6, 15) = 3$$

$$2t \equiv 1 \pmod{5}$$

$$5 = 2 \times 2 + 1 \quad 1 = 5 - 2 \times 2$$

$$2 = 2 \times 1 + 0 \quad = 5 + (-2) \times 2$$

$$\text{inverse: } -2+k$$

$$-2+5 = 3$$

$$3 \times 2t \equiv 3 \times 1 \pmod{5}$$

$$t \equiv 3 \pmod{5}, 5k+3 = t, k \in \mathbb{Z}$$

$$\text{Ans: } 3+11k, k \in \mathbb{Z} \text{ or}$$

$$6+11k, k \in \mathbb{Z}$$

$$x = 6(5k+3) + 5 \quad x \equiv 3 \pmod{10}$$

$$= 30k + 23$$

$$x = 10u + 3, u \in \mathbb{Z}$$

$$x \equiv 23 \pmod{30}$$

$$10u + 3 \equiv 23 \pmod{30}$$

$$10u \equiv 20 \pmod{30} \rightarrow u \equiv 2 \pmod{3}$$

$$\gcd(10, 30) = 10 \quad u = 3n + 2, n \in \mathbb{Z}$$

$$x = 10(3n+2) + 3$$

$$= 30n + 23$$

$$x \equiv 23 \pmod{30}$$

$$\text{Ans: } x = 30n + 23, n \in \mathbb{Z} \checkmark$$

8.

$$(a) P(n) : 1 \times 2 + 2 \times 3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

$$P(1) : 1 \times 2 = 2$$

$$\frac{1 \times (1+1)(1+2)}{3} = \frac{1 \times 2 \times 3}{3} = 2, \quad P(1) \text{ is true since } 1 \times 2 = \frac{1 \times 2 \times 3}{3} = 2$$

Assume $P(k)$ is true:

$$1 \times 2 + 2 \times 3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$$

$P(k+1)$:

$$1 \times 2 + 2 \times 3 + \dots + k(k+1) + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3}$$

$$= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} = \frac{(k+1)(k+2)(k+3)}{3} = \frac{(k+1)[(k+1)+1][(k+1)+2]}{3}$$

$$\Rightarrow P(k+1) \text{ is also true, hence, } 1 \times 2 + 2 \times 3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

(b) Find the **flaw** with the following "proof" that $a^n = 1$ for all nonnegative integers n , whenever a is a nonzero real number.

Basic Step: $a^0 = 1$ is true by the definition of a^0 .

Inductive Step: Assume that $a^j = 1$, for all nonnegative integers j with $j \leq k$.

Then we can get:
$$a^{k+1} = \frac{a^k \times a^k}{a^{k-1}} = \frac{1 \times 1}{1} = 1$$

在算式 $a^{k+1} = \frac{a^k \times a^k}{a^{k-1}}$ 中, 成立條件需是 $a^{k-1} = a^k = a^{k+1} = 1$,

所以 $a^1 = 1$ 、 $a^2 = 1$ 也應在 Basic Step 一併證明, 而不是僅

證明 $a^0 = 1$ 。 $a^j = 1$ 也是錯誤的, $a^j = 1$ 不代表 a^{k+1} 也 $= 1$, 除了

$a=1$ 以外, 對於其他非零實數並不成立。

(Sol.)

The flaw comes in the inductive step, where we are implicitly assuming that $k \geq 1$ in order to talk about a^{k-1} in the denominator (otherwise the exponent is not a nonnegative integer, so we cannot apply the inductive hypothesis).

$$a^{k+1} = \frac{a^k \cdot a^k}{a^{k-1}} = \frac{1 \cdot 1}{1} = 1$$

Our basis step was $n=0$, so we are not justified in assuming that $k \geq 1$ when we try to prove the statement for $k+1$ in the inductive step. Indeed, it is precisely at $n=1$ that the proposition breaks down.